1. If $2x^y + 3y^x = 20$, then $\frac{dy}{dx}$ at $(2, 2)$ is equal to:
[2023 (06 Apr Shift 1)]
(1) $-\left(\frac{2+\log_e 8}{3+\log_e 4}\right)$
(2) $-\left(\frac{3+\log_e 16}{4+\log_e 8}\right)$
(3) $-\left(\frac{3+\log_e 8}{2+\log_e 4}\right)$
(4) $-\left(\frac{3+\log_e 4}{2+\log_e 8}\right)$

2. Let $f(x)=\frac{\sin x+\cos-\sqrt{2}}{\sin\ x-\cos\ x}, x \in[0,\ \pi]-\left\{\frac{\pi}{4}\right\}$, then $f\left(\frac{7\pi}{12}\right)f''\left(\frac{7\pi}{12}\right)$ is equal to
[2023 (08 Apr Shift 1)]
(1) $\frac{2}{9}$
(2) $\frac{-2}{3}$
(3) $\frac{-1}{3\sqrt{3}}$
(4) $\frac{2}{3\sqrt{3}}$

3. Let k and m be positive real numbers such that the function $f(x)=\begin{cases} 3x^2+k\sqrt{x+1}, & 0<x<1 \\ mx^2+k^2, & x\geq 1 \end{cases}$ is differentiable for all $x>0$. Then $\frac{8f'(8)}{f'\left(\frac{1}{8}\right)}$ is equal to
[2023 (08 Apr Shift 2)]

ANSWER KEYS

1. (1) **2.** (1) **3.** (309)

1. (1)

Given,

$2x^y + 3y^x = 20$

Now differentiating both sides w.r.t x, we get

$2x^y\left(y' \ln x + \frac{y}{x}\right) + 3y^x\left(\ln y + \frac{x}{y} \cdot y'\right) = 0$ {here $\ln = \log_e$}

Putting $x = 2$ & $y = 2$, we get

$\Rightarrow 8(\ln 2 \cdot y' + 1) + 12(\ln y + y') = 0$

$\Rightarrow 8(\ln 2 \cdot y' + 1) + 12(\ln y + y') = 0$

$\Rightarrow y' = -\left(\frac{3\ln 2+2}{2\ln 2+3}\right)$

$\Rightarrow y' = -\left(\frac{2+\ln 8}{3+\ln 4}\right)$

2. (1)

Give that:

$f(x) = \frac{\sin x + \cos x - \sqrt{2}}{\sin x - \cos x}$

Convert the numerator and denominator in the form of $\sin(A \pm B)$.

$\Rightarrow f(x) = \frac{\sqrt{2}s\text{in}\left(x+\frac{\pi}{4}\right) - \sqrt{2}}{\sqrt{2}\left(\sin\left(x-\frac{\pi}{4}\right)\right)}$

$\Rightarrow f(x) = \frac{\sin\left(x+\frac{\pi}{4}\right) - 1}{\sin\left(x-\frac{\pi}{4}\right)}$

$\Rightarrow f\left(x + \frac{\pi}{4}\right) = \frac{\cos\ x - 1}{\sin\ x}$

$\Rightarrow f\left(x + \frac{\pi}{4}\right) = -\tan\frac{x}{2}$

$\Rightarrow f(x) = -\tan\left(\frac{x}{2} - \frac{\pi}{8}\right)$

$\Rightarrow f'(x) = \frac{-1}{2}\sec^2\left(\frac{x}{2} - \frac{\pi}{8}\right)$

$\Rightarrow f''(x) = -\frac{1}{2}\left(\sec^2\left(\frac{x}{2} - \frac{\pi}{8}\right)\tan\left(\frac{x}{2} - \frac{\pi}{8}\right)\right)$

$\Rightarrow f\left(\frac{7\pi}{12}\right) = -\tan\left(\frac{7\pi}{24} - \frac{\pi}{8}\right) = -\tan\left(\frac{4\pi}{24}\right)$

$\Rightarrow -\tan\left(\frac{\pi}{6}\right) = -\frac{1}{\sqrt{3}}$

Also,

$\Rightarrow f''\left(\frac{7\pi}{12}\right) = -\frac{1}{2} \times \left(\frac{2}{\sqrt{3}}\right)^2 \times \frac{1}{\sqrt{3}} = -\frac{2}{3\sqrt{3}}$

$\Rightarrow f\left(\frac{7\pi}{12}\right)f''\left(\frac{7\pi}{12}\right) = \frac{2}{9}$

Hence this is the correct option.

3. (309)

Since, $f(x)=\begin{cases} 3x^2+k\sqrt{x+1}; & 0<x<1 \\ mx^2+k^2; & x\geq 1 \end{cases}$ is differentiable at $x=1$, so function must be continuous at $x=1$, hence

LHL $=$ RHL

$3+k\sqrt{2}=m+k^2$

$\Rightarrow k^2-k\sqrt{2}+m-3=0 \quad \ldots.(1)$

And,

$f'(x)=\begin{cases} 6x+\frac{k}{2\sqrt{x+1}}; & 0<x<1 \\ 2mx; & x>1 \end{cases}$

So,

$f'(1^-)=f'(1^+)$

$\Rightarrow 6+\frac{k}{2\sqrt{2}}=2m$

$\Rightarrow m=3+\frac{k}{4\sqrt{2}} \quad \ldots(2)$

Putting in (1), we get

$k^2-k\sqrt{2}+3+\frac{k}{4\sqrt{2}}-3=0$

$\Rightarrow k^2-\left(\sqrt{2}-\frac{1}{4\sqrt{2}}\right)k=0$

$\Rightarrow k^2-\left(\frac{7\sqrt{2}}{8}\right)k=0$

$\Rightarrow k\left(k-\frac{7\sqrt{2}}{8}\right)=0$

$\Rightarrow k=\frac{7\sqrt{2}}{8}$

So,

$m=1+\frac{k}{12\sqrt{2}}=1+\frac{7\sqrt{2}}{96\sqrt{2}}=\frac{103}{96}$

Hence,

$f'(x)=\begin{cases} 6x+\frac{7\sqrt{2}}{16\sqrt{x+1}}; & 0<x<1 \\ \frac{103x}{16}; & x>1 \end{cases}$

So,

$\frac{f'(8)}{f'\left(\frac{1}{8}\right)}=\frac{103\times 8}{16}\times\frac{1}{\left(\frac{6}{8}+\frac{7\sqrt{2}}{16\sqrt{\frac{9}{8}}}\right)}$

$\Rightarrow \frac{f'(8)}{f'\left(\frac{1}{8}\right)}=\frac{103}{2}\times\frac{3}{4}$

$\Rightarrow \frac{8f'(8)}{f'\left(\frac{1}{8}\right)}=309$